



Dow Packaging & Specialty Plastics

Product Data Sheet

SURLYN™ 1650

Ionomer

General Information

Product Description	SURLYN™ 1650 is an ionomer of ethylene acid copolymer. The resin can be processed in conventional blown film, cast film, sheet extrusion and coextrusion equipment designed to process polyethylene and ethylene copolymer type resins.
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Status

Material Status	Commercial: Active
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Typical Characteristics

Composition	Zinc Ionomer
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Typical Properties

Physical	Nominal Values	Test Method(s)	
*Density ()	0.94 g/cm ³	ASTM D792	ISO 1183
*Melt Flow Index (190°C/2.16kg)	1.8 g/10 min	ASTM D1238	ISO 1133
Thermal	Nominal Values	Test Method(s)	
*Melting Point (DSC)	97 °C (206.6 °F)	ASTM D3417	ISO 3146
Freezing Point (DSC)	73 °C (163.4 °F)	ASTM D3417	ISO 3146
Vicat Softening Point ()	73 °C (163.4 °F)	ASTM D1525	ISO 306

Processing Information

*Maximum Processing Temperature	285 °C (545 °F)
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General Processing Information	SURLYN™ 1650 is normally processed at melt temperatures ranging from 160°-260°C (320°-500°F) in blown and flat die equipment. Typical extruder profiles are shown below. Actual processing temperatures will usually be determined by either the specific equipment or substrate or one of the other polymers in a coextrusion. Materials of construction used in the processing of this resin should be corrosion resistant. Stainless steels of the types 316, 15-5PH, and 17-4PH are excellent, as is quality chrome or nickel plating, and in particular duplex chrome plating. Type 410 stainless steel is satisfactory, but needs to be tempered at a minimum temperature of 600°C (1112°F) to avoid hydrogen-assisted stress corrosion cracking. Alloy steels such as 4140 are borderline in performance. Carbon steels are not satisfactory. While stainless steels can provide adequate corrosion protection, in some cases severe purging difficulties have been encountered. Nickel plating has been satisfactory, but experiments have shown that chrome surfaces have the least adhesion to acid based polymers. In recent years, the quality of chrome plating has been deteriorating due to environmental pressures, and the corrosion protection has not always been adequate. Chrome over top of stainless steel seems to provide the best combination for corrosion protection and ease of purging. If surface properties of the extruded resin require modification (such as, lower C.o.F. for packaging machine processing), refer to the CONPOL™ Processing Additive Resins product information guide. After processing SURLYN™, purge the material out using a polyethylene resin, preferably with a lower melt flow rate than the SURLYN™ resin in use. The "Disco Purge Method" is suggested as the preferred purging method, as this
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method usually results in a more effective purging process. Information on the Disco Purge Method can be obtained via your Dow Sales Representative.

Never shut down the extrusion system with SURLYN™ in the extruder and die. Properly purge out the SURLYN™ with a polyethylene, and shut down the line with polyethylene or polypropylene in the system.

Blown Film	Nominal Values
Processing Information	A suggested initial extruder temperature set profile.
Feed Zone	135 °C (275 °F)
Second Zone	160 °C (320 °F)
Third Zone	185 °C (365 °F)
Fourth Zone	185 °C (365 °F)
Fifth Zone	185 °C (365 °F)
Adapter Zone	185 °C (365 °F)
Die Zone	185 °C (365 °F)

Cast Film / Sheet	Nominal Values
Processing Information	A suggested initial extruder temperature set profile.
Feed Zone	160 °C (320 °F)
Second Zone	210 °C (410 °F)
Third Zone	235 °C (455 °F)
Fourth Zone	235 °C (455 °F)
Fifth Zone	235 °C (455 °F)
Adapter Zone	235 °C (455 °F)
Die Zone	235 °C (455 °F)

FDA Status Information	SURLYN™ 1650 complies with Food and Drug Administration Regulation 21 CFR 177.1330(a) - - Ionomeric resins, subject to the limitations and requirements therein. This Regulation describes polymers that may be used in contact with food, subject to the finished food-contact article meeting the extractive limitations under the intended conditions of use, as shown in paragraph (c) of the Regulation.
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Regulatory Information	For information on regulatory compliance outside of the U.S.A., consult your local Dow representative.
Safety & Handling	For information on appropriate Handling & Storage of this polymeric resin, please refer to the material Safety Data Sheet.

A Product Safety Bulletin, material Safety Data Sheet, and/or more detailed information on extrusion processing and/or compounding of this polymeric resin for specific applications are available from your Dow representative.

Product Stewardship

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<https://www.dow.com/en-us/support/product-safety.html>

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http://www.dow.com/products_services

Revision Date: 26-February-2021

P&SP Disclaimer

Additional Information

To contact Dow via Toll-Free or Local Toll phone numbers in specific countries, please see the following webpage:

<https://www.dow.com/en-us/support/contact-representative.html>

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Updated February 2021

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Version: 154.0

Last modified at 8/11/2019 9:42 AM